

RhinoRoTex

VORTEX FLOW CONTROL CHAMBER

ROTEX600 - ROTEX1200 | DATA SHEET

OVERVIEW

RhinoRoTex is the vortex flow control chamber in the RHINO range, sold as ROTEX600 to ROTEX1200. A passive vortex regulator on the outlet holds stormwater discharge to a pre-set maximum rate, protecting downstream drainage from overload during heavy rainfall.

The unit fits curved or flat surfaces as standard. It is supplied factory-fitted in an HDPE twinwall chamber, arriving on site ready for connection and backfill with no benching, or as a standalone unit for fitting into an existing concrete chamber.

600

ROTEX600

750

ROTEX750

900

ROTEX900

1050

ROTEX1050

1200

ROTEX1200

HOW THE VORTEX WORKS

Swirling water does the regulating, so there are no moving parts. Stormwater entering the flow controller is set rotating; as inflow rises the vortex strengthens and centrifugal forces restrict the flow, holding the outflow to a pre-set maximum rate even as storm conditions vary. The excess is held back in the chamber and upstream storage and released as the storm passes.

DESIGN COMPLIANCE

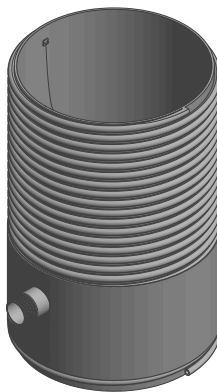
- BS EN 13598-2 (2009) plastic inspection chambers
- Sewers for Adoption 7th Edition (SfA7)
- Design and Construction Guidance (DCG), Type D and E chambers
- Building Regulations Part H1
- Adoptable and non-adoptable applications

CHAMBER OPTIONS

Pipework in 160 EN1401 or 150 to 750 twinwall, depending on chamber size. An optional ladder is made to suit the chamber height. Inlet orientation depends on the chosen pipework configuration and diameter.

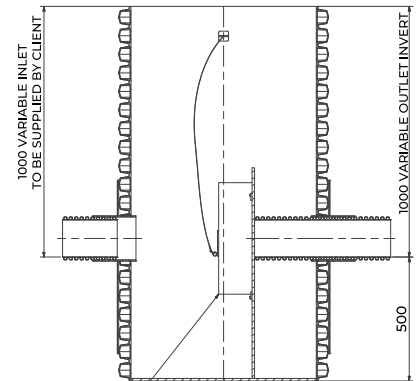
INSTALLATION

Installation design should consider traffic loading, ground conditions and variable groundwater levels. If in any doubt consult the scheme designer or engineer. A suitably qualified professional is required to sign off the installation.



ROTEX1200 factory-fitted chamber.

SuDS
enviro
BESPOKE, STANDARDISED



FLOW CONTROLLER DESIGNED TO SUIT SITE CONDITIONS

ROTEX750 section A-A. Dimensions in mm from drawing RoTex-C750-1.5.

NOT TO SCALE

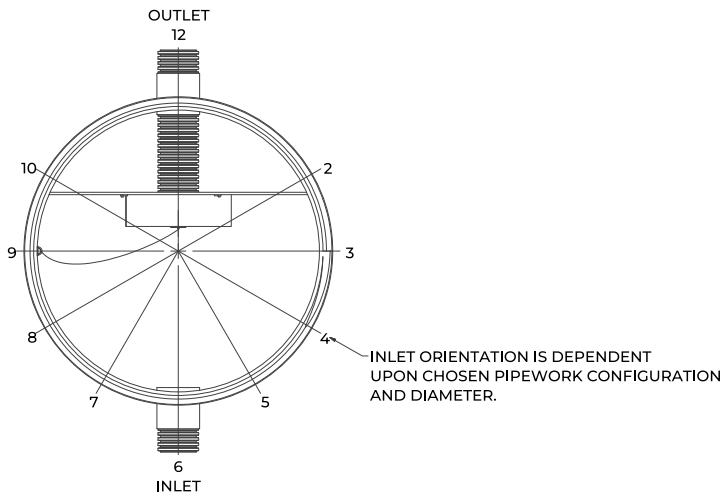
FEATURES & BENEFITS

- Passive vortex control, no power supply
- Flow controller designed to suit the site
- Factory-fitted in HDPE, no benching
- Standalone unit for concrete chambers
- Fits curved or flat surfaces as standard
- Emergency bypass opened from the surface
- Housed in 600 to 1200 mm chambers
- No moving parts, low maintenance

SPECIFICATION

Sizes	ROTEX600, 750, 900, 1050, 1200 (Ø600 to 1200 mm)
Supply	Factory-fitted HDPE chamber, or standalone unit for concrete chambers
Control	Passive vortex regulator, pre-set discharge rate
Sump	300 mm (600), 500 mm (750 to 1200) as drawn
Depth	Adoptable up to 2000 mm, non-adoptable up to 3000 mm
Design input	Design head (m) and discharge rate (L/s)

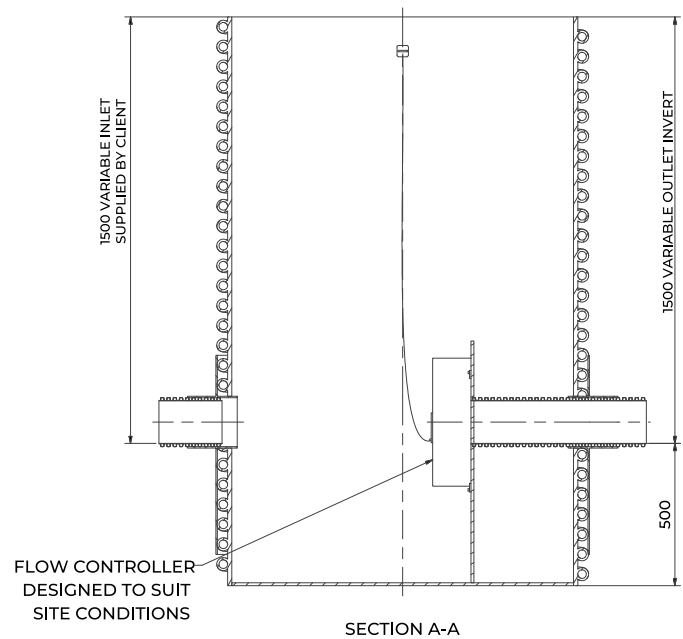
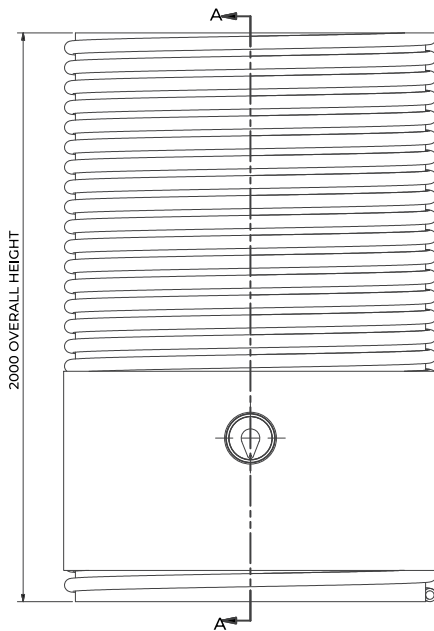
PLAN VIEW | OUTLET AT 12, INLET AT 6



ROTEX-C1200-2

Chamber	Ø1200 mm HDPE twinwall
Overall height	2000 mm
Inlet	6 o'clock, invert supplied by client (1500 mm drawn)
Outlet	12 o'clock, 1500 mm variable invert
Sump	500 mm
Pipework	160 EN1401, or 150, 225, 300, 450, 600, 750 twinwall
Access	Optional ladder to suit chamber height

FRONT ELEVATION AND SECTION A-A



READING THE DRAWING

Drawing RoTex-C1200-2 shows a Ø1200 chamber with a 2 m overall height. Invert levels are measured down from the top of the chamber; the inlet invert is supplied by the client and the outlet invert is made to suit.

The flow controller sits on the outlet, 500 mm above the base, leaving the sump below it.

Each chamber size is made in a family of product codes, one per pipework size, for example RoTex-C1200150-2 for 150 twinwall. The flow controller is designed to suit the site conditions.

NOT TO SCALE Reproduced from the SuDS Enviro drawing pack for illustration. Dimensions in millimetres. Confirm project dimensions against the order drawing.

1 VORTEX CONTROL

The flow controller is mounted on the outlet. Water entering it is set rotating, and the vortex strengthens as the inflow rises, so centrifugal forces restrict the flow and the discharge stays at or below the pre-set rate. There is no external power and nothing mechanical to fail.

2 STORAGE AND SUMP

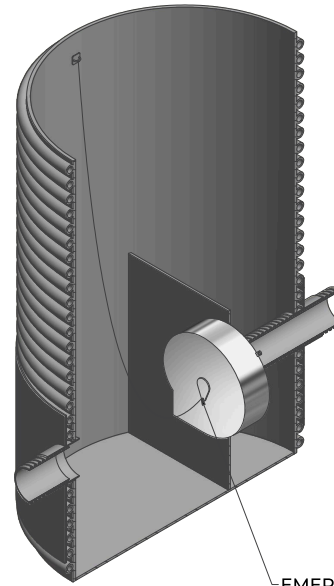
Inflow above the controlled rate backs up in the chamber and the upstream attenuation, then drains down as the storm passes. The sump below the outlet collects silt and debris so they can be removed during routine maintenance.

3 EMERGENCY BYPASS

An emergency bypass door on the flow controller is opened by a cable from the surface. The chamber can be drained for inspection, or a blockage cleared, without anyone entering it.

4 FACTORY-FITTED OR STANDALONE

The factory-fitted option arrives as a complete system ready for connection and backfill, removing the cost and hazard of in-ground benching. The standalone unit mounts on the curved or flat wall of an existing concrete chamber.



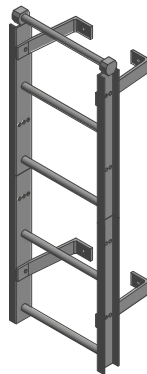
EMERGENCY BYPASS DOOR
WITH CABLE OPENING
FROM SURFACE

ROTEX1200 cutaway. Drawing RoTex-C1200-2.

DESIGN HEAD

The design head is the top water level of the upstream storage minus the invert level of the outlet pipe. For oversized storage pipes, the top water level is the upstream invert plus the storage pipe diameter.

Send the design head with the permitted discharge rate and the vortex is sized to suit.



OPTIONAL LADDER TO SUIT
CHAMBER HEIGHT

PIPEWORK CODES | ROTEX1200

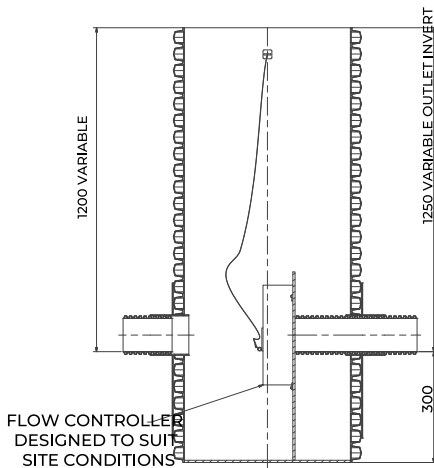
160 EN1401	RoTex-C1200160-2
150 twinwall	RoTex-C1200150-2
225 twinwall	RoTex-C1200225-2
300 twinwall	RoTex-C1200300-2
450 twinwall	RoTex-C1200450-2
600 twinwall	RoTex-C1200600-2
750 twinwall	RoTex-C1200750-2

From drawing RoTex-C1200-2. Smaller chambers take a shorter list; see page 4.

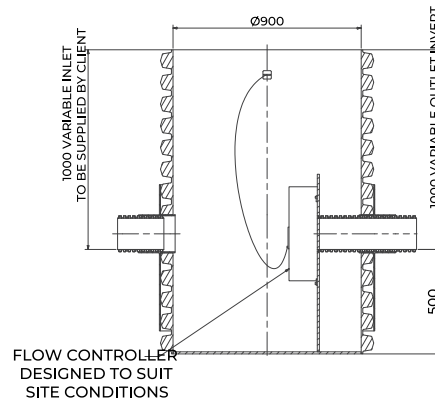
SIZING INPUTS

- Design head in metres
- Permitted discharge rate in L/s
- Chamber size: ROTEX600 to ROTEX1200
- Inlet and outlet pipe size and type
- Inlet clock positions
- Invert depths and cover level
- Adoptable or non-adoptable
- Factory-fitted or standalone unit

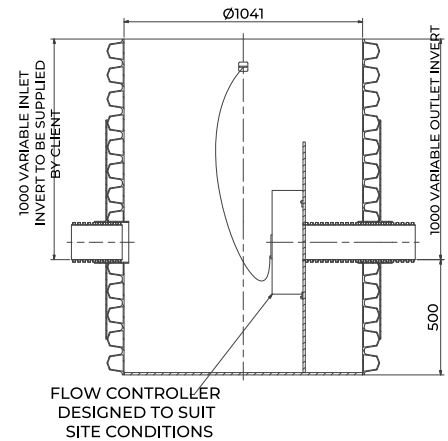
Our technical team returns a unit with the flow controller designed to suit the site.



ROTEX600 - drawing RoTex-C600-1.5



ROTEX900 - drawing RoTex-C900-1.5



ROTEX1050 - drawing RoTex-C1050-1.5

NOT TO SCALE Sections are shown at different scales so each fits the column. Each is one drawn configuration; overall height, pipework and inverts are made to order.

CODE	CHAMBER	PIPEWORK OPTIONS	OVERALL HEIGHT	INLET INVERT	OUTLET INVERT	SUMP
ROTEX600	Ø600 mm	160 EN1401, 150 / 225 / 300 twinwall	1500 mm	1200 mm	1250 mm	300 mm
ROTEX750	Ø750 mm	160 EN1401, 150 / 225 / 300 twinwall	1500 mm	1000 mm	1000 mm	500 mm
ROTEX900	Ø900 mm	160 EN1401, 150 to 450 twinwall	1500 mm	1000 mm	1000 mm	500 mm
ROTEX1050	Ø1050 mm	160 EN1401, 150 to 600 twinwall	1500 mm	1000 mm	1000 mm	500 mm
ROTEX1200	Ø1200 mm	160 EN1401, 150 to 750 twinwall	2000 mm	1500 mm	1500 mm	500 mm

Figures as drawn on RoTex-C600-1.5, C750-1.5, C900-1.5, C1050-1.5 and C1200-2. Inverts are variable and measured down from the top of the chamber; the inlet invert is supplied by the client.

INSTALLATION

- Bed and surround to suit the traffic loading and ground conditions
- Allow for variable groundwater levels and flotation
- Set the outlet to the design invert; the flow controller is factory-fitted
- Keep the bypass cable free and reachable from cover level
- Have the installation signed off by a suitably qualified professional

COMPLIANCE

- BS EN 13598-2 (2009)
- SfA7 (Sewers for Adoption)
- DCG Type D and E chambers
- Building Regulations Part H1
- Adoptable and non-adoptable applications

SPECIFY RHINOROTEX

Send us the design head and discharge rate and our team will return a unit designed for the site.

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